

MEMVERSE

The zero-trust memory firewall for AI agents.

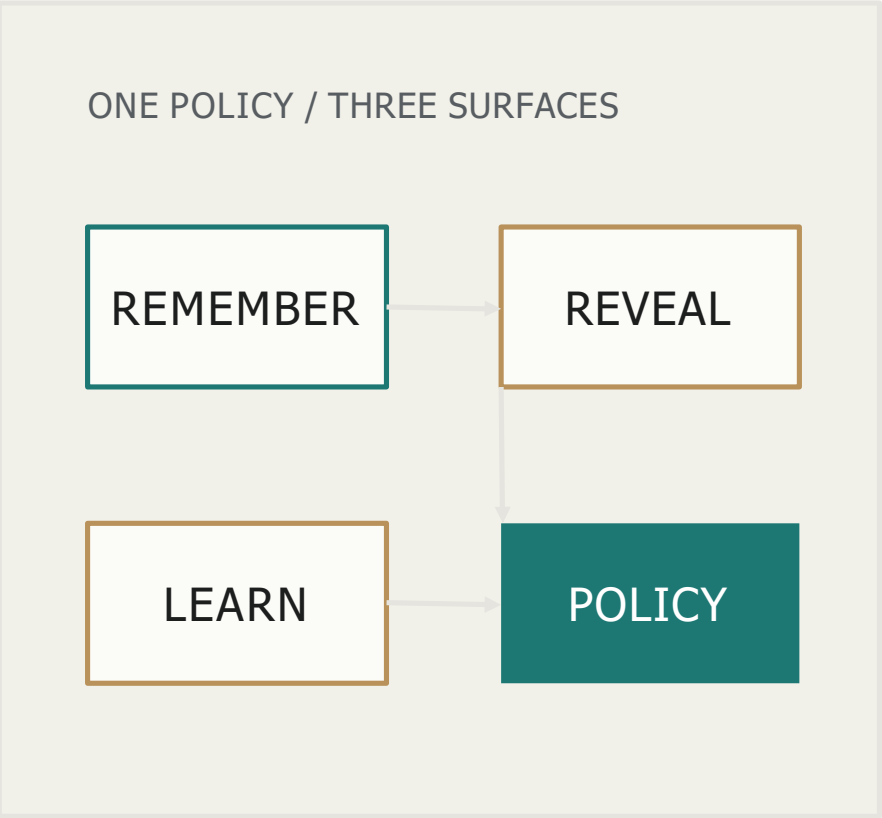
PRIMARY THEME

Cybersecurity

Memory that remembers the rule, not just the data.

TEAM MEMBERS

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AI is becoming persistent; control is still fragmented

01 REMEMBER

A conversation or derived fact can become persistent memory.

The user needs retention, expiry, edit, and delete to be visible policy decisions.

02 REVEAL

Raw context may leave the device or workspace for an external model or tool.

The user needs field-level suppress, generalize, redact, or release decisions.

03 LEARN

Data can influence a model update, retrieval index, analytics system, or shared model.

The user needs provenance and a rule for training or reuse.

THE GAP

Users can choose what to tell AI — but not a portable policy for what AI may retain, reveal, or learn.

The problem has three distinct control surfaces

They are connected, but not interchangeable: storage controls, transmission controls, and learning controls answer different questions.

Surface	What can happen	What the user needs
Remember	A conversation or derived fact becomes persistent memory.	Visible retention, expiry, edit, and delete policy.
Reveal	Raw context leaves the device or workspace for an external model or tool.	Field-level suppress, generalize, redact, or release decision.
Learn	Data influences a model update, retrieval index, analytics system, or shared model.	Provenance plus an explicit rule for training or reuse.

Deleting a memory does not prove that a model has forgotten a training example. Keeping data on-device does not make every update non-sensitive.

Existing tools protect slices; lifecycle gaps remain

The opportunity is a zero-trust control layer around every memory event — not another memory database.

Existing capability	What it does well	Gap MEMVERSE targets
Prompt redaction	Removes obvious identifiers before a request.	Does not govern persistence, retrieval, or future reuse.
Memory infrastructure	Stores, searches, compresses, and audits agent memory. [10] [11]	Usually optimizes recall; policy is not re-evaluated at every event.
Federated learning	Keeps raw training data local while aggregating updates. [4]	Update content and release conditions still need governance.
Secure aggregation / PETs	Can hide individual contributions under a threat model. [5]	Does not decide whether the contribution should exist or expire.

MEMVERSE: policy is checked before write, read, reveal, and reuse.
It intervenes before data crosses a boundary — not only after the fact.

USP: a zero-trust firewall for the AI memory lifecycle

Every memory event is evaluated before it can influence the agent again.

01

Classify risk

Detect sensitive data, suspicious instructions, cross-user references, and high-impact actions.

02

Evaluate intent

Check purpose, consent, actor, destination, retention, and policy version.

03

Enforce one of four actions

Block, transform, local-only, or allow-with-expiry. Revoked Passport = fail closed.

04

Prove the decision

Attach a Memory Passport and record a human-readable, tamper-evident receipt.

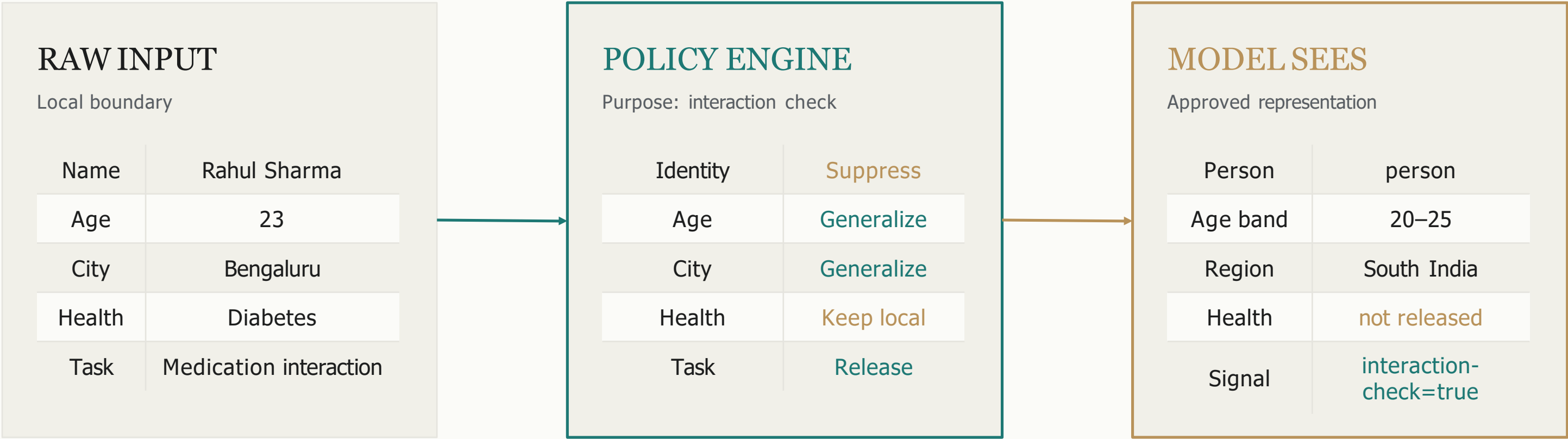
MEMORY PASSPORT

sensitivity · purpose · consent · destination · TTL · integrity hash · policy version · revocation state

NIST mapping: the policy engine decides; the gateway enforces at the AI memory and agent-data boundary. [9]

The same input can produce different safe representations

Illustrative policy behavior: release the task signal, not the raw identity.



LIMITATION: generalization can reduce direct identifiability, but it is not a universal anonymity guarantee. The MVP will test linkage and re-identification on a controlled benchmark.

The Memory Passport makes control revocable and inspectable

The policy is checked when memory is written, when it is read, and when consent or purpose changes.

01 / WRITE

Screen before persistence Detect sensitive or malicious content, then block, transform, or attach a Passport before storing.

02 / READ

Re-evaluate before release Check the current actor, purpose, destination, and Passport state before the agent receives context.

03 / REVOKE

Fail closed after change When consent or purpose changes, future retrieval and outbound release are denied; deletion is queued.

RECEIPT VIEW

What the system can prove

event ID and timestamp
purpose and destination
fields transformed
TTL and policy version
integrity hash
revocation status

Human-readable and replayable.

WHY IT MATTERS: OWASP recommends validation, isolation, expiry, sensitive-data audits, and integrity checks before agent memory is persisted. [7]

MVP features prove the security loop end to end

The demo is a blocked, transformed, revoked, or receipted memory event — not just a dashboard.

ENFORCE

Policy-as-code gateway

OPA / Rego or typed JSON rules. Proof: repeatable decision for the same event and policy.

DETECT

Sensitive-data detector

Presidio, regex, and human override.
Proof: risk fields appear with a reason.

DEFEND

Memory-poisoning signal

Flag hidden instructions and unsafe tool requests. Proof: suspicious memory is quarantined.

CONTROL

Memory Passport

Purpose, consent, TTL, and revocation.
Proof: retrieval fails after revoke.

TRANSFORM

Router and transformer

Suppress, redact, generalize, tokenize, or keep local. Proof: only approved payload exits.

PROVE

Tamper-evident receipts

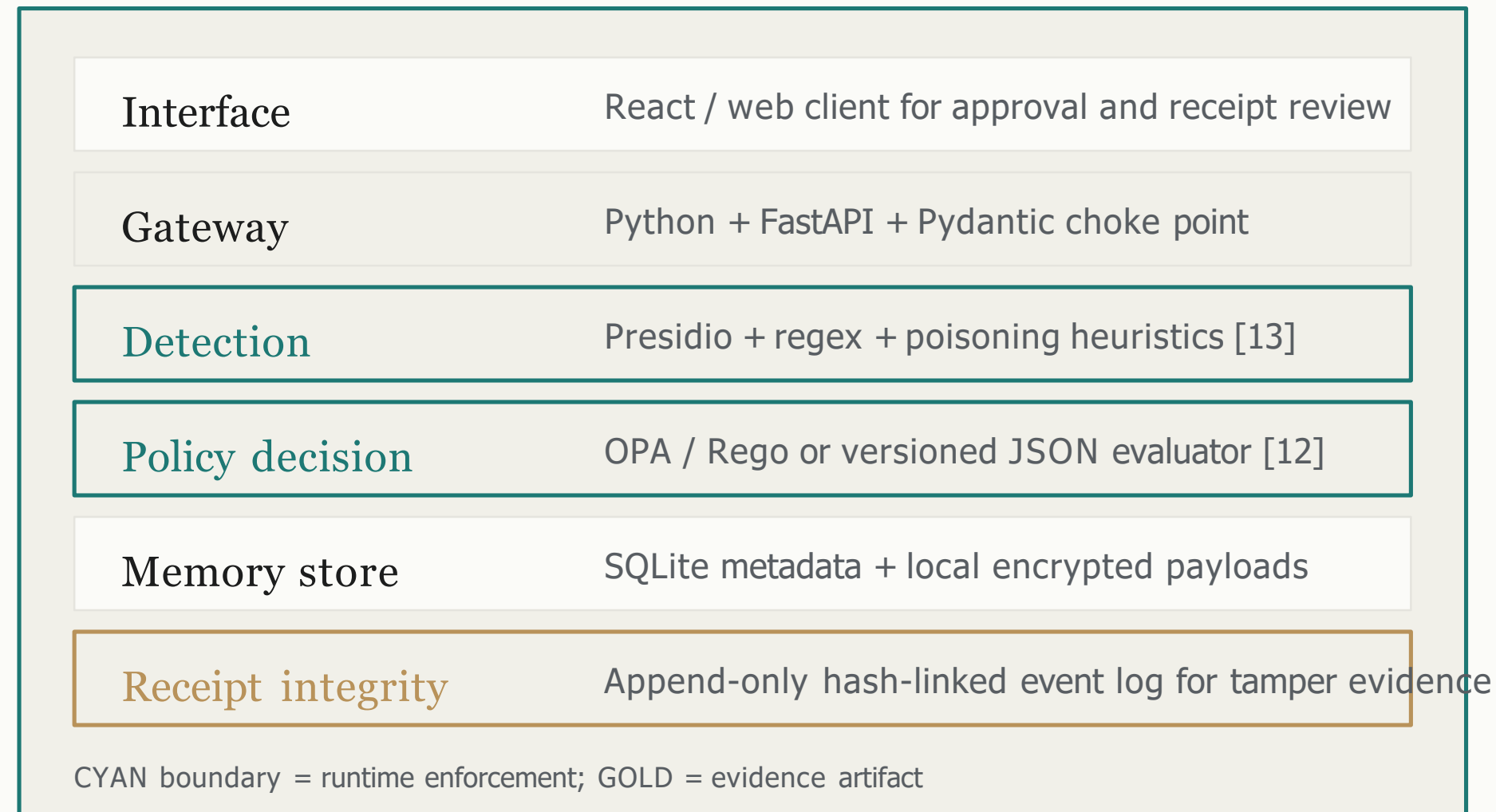
Hash-linked log of policy events. Proof: every allow, block, transform, and revoke is inspectable.

SCOPE DISCIPLINE

Prove runtime memory governance first. Federated learning and secure aggregation are extension paths, not MVP claims.

The stack is open, modular, and realistic for a hackathon

Open components handle detection and policy; MEMVERSE contributes the lifecycle contract and enforcement logic.



WHY THIS IS FEASIBLE

OPA decouples policy decisions from enforcement and accepts structured JSON input. [12]

Presidio supplies extensible PII signals, but warns that automated detection is not complete. [13]

The gateway owns the critical boundary: model calls, memory writes, retrieval, and tool requests.

Security claim = bounded, inspectable, testable.

The pitch maps directly to the Round 1 evaluation rubric

Every criterion has a visible argument, a bounded implementation, and a proposed proof point.

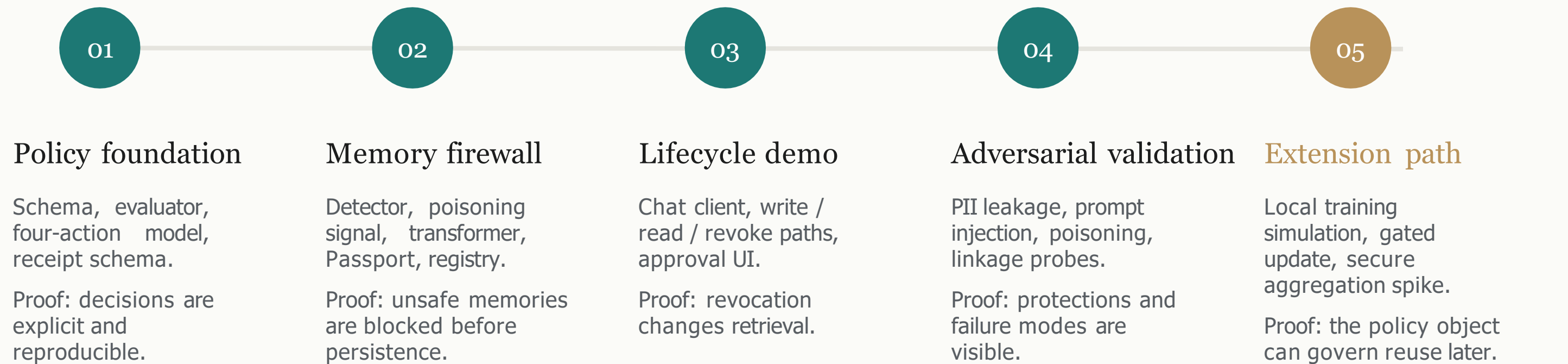
Criterion	What judges should see	Proof in the deck / demo
Innovation and originality	Policy-carrying memory, retrieval-time revocation, and defense against leakage plus poisoning.	Memory Passport, quarantine flow, and tamper-evident receipt.
Feasibility	Open components, narrow MVP boundary, modular gateway, reproducible policies.	OPA / FastAPI / Presidio stack and five-phase roadmap.
Problem-solving approach	Separate Remember, Reveal, and Learn events; enforce at each boundary.	Lifecycle flow and field-level transformation example.
Clarity of presentation	One vocabulary: Memory Passport, policy gateway, receipt, four actions.	Consistent diagrams, tables, and explicit limitations.
Potential impact	Reusable across cybersecurity and high-consequence application tracks.	Theme-fit slide and focused market wedge.

ACCEPTANCE TESTS

- zero prohibited fields in egress · revocation blocks future retrieval
- poisoning probes quarantine · utility plus p50 / p95 latency are reported

The build plan turns the USP into evidence

First prove the memory firewall; only then extend the same policy object toward collaborative learning.



CURRENT STATUS / NEXT PROOF

Concept and presentation prototype → live demo of block, transform, quarantine, revoke, retrieval denial, and receipt.

One policy engine can serve five high-impact pathways

Cybersecurity is the wedge.
Privacy-preserving AI is the scale.

MEMVERSE gives agent builders a zero-trust boundary for memory, tools, APIs, and model context.
Personalization can continue without uncontrolled persistence.

MARKET WEDGE

Developers and organizations deploying memory-enabled AI agents in regulated or high-impact workflows.

THEME FIT / APPLICATION PATH	
Cybersecurity	Primary fit: secure memory, tools, APIs, and agent context.
Healthcare & MedTech	Purpose-limited health context with receipts and revocation.
FinTech	Selective release of identity, account, and transaction fields.
Women Safety	Expire, local-only, or human-approve high-risk narratives.
Open Innovation	One policy contract across models and organizations.
EVIDENCE OWASP [7][8] · NIST [1][3][9] · OPA [12] · Presidio [13]	

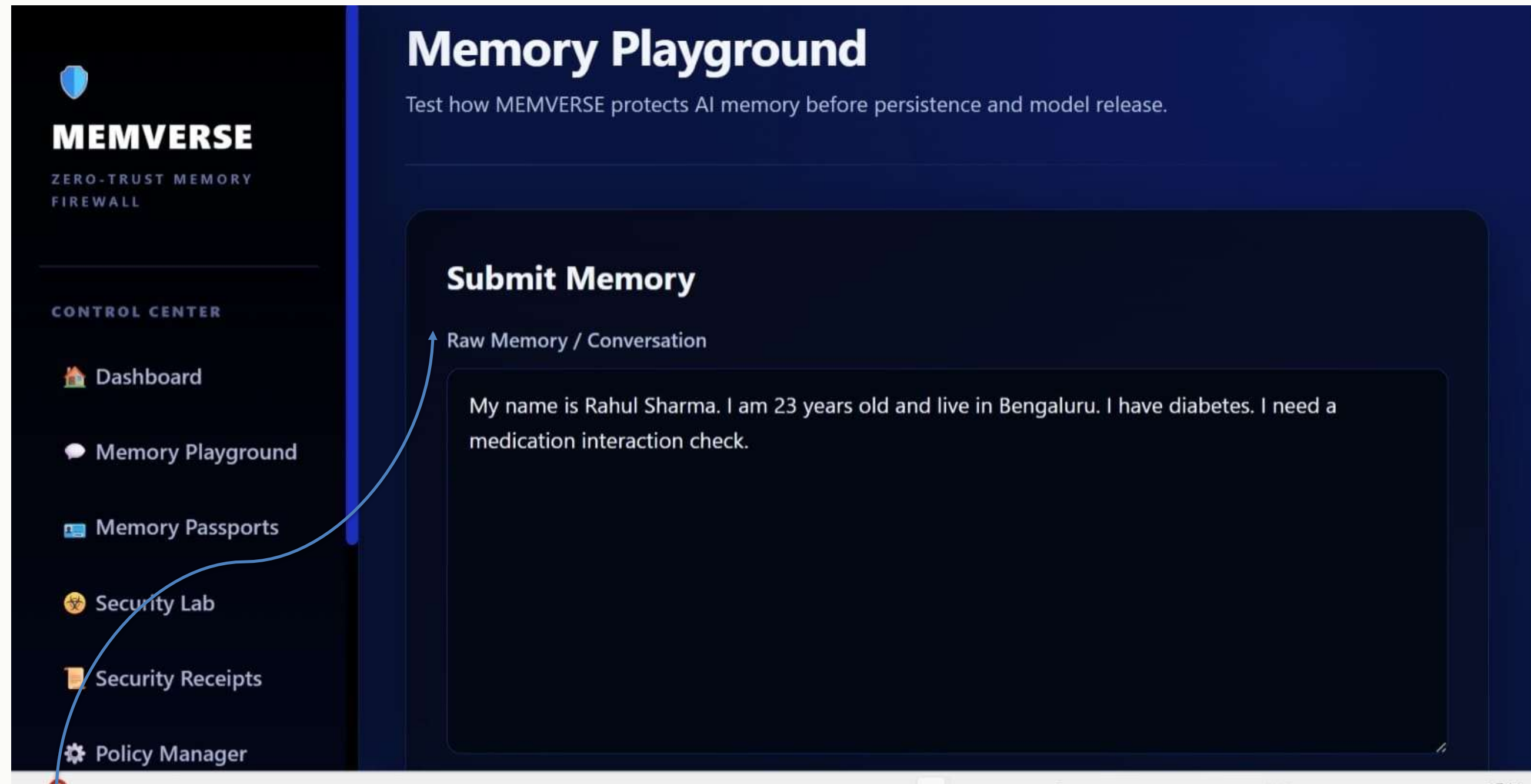
NEXT PROOF: block a memory, revoke its Passport, deny retrieval, and show the signed decision receipt.

Security Dashboard – Real-time memory lifecycle overview



Shows total memories and their status. Policy engine continuously evaluates each memory event before persistence or release.

Memory Playground – Submit raw memory



User enters personal/contextual data. Policy engine applies suppression, generalization, or transformation before persistence.

Policy Gateway – Consent & destination check

CONTROL CENTER

Dashboard

Memory Playground

Memory Passports

Security Lab

Security Receipts

Policy Manager

● SYSTEM PROTECTED

Policy Engine: Online

Gateway: Online

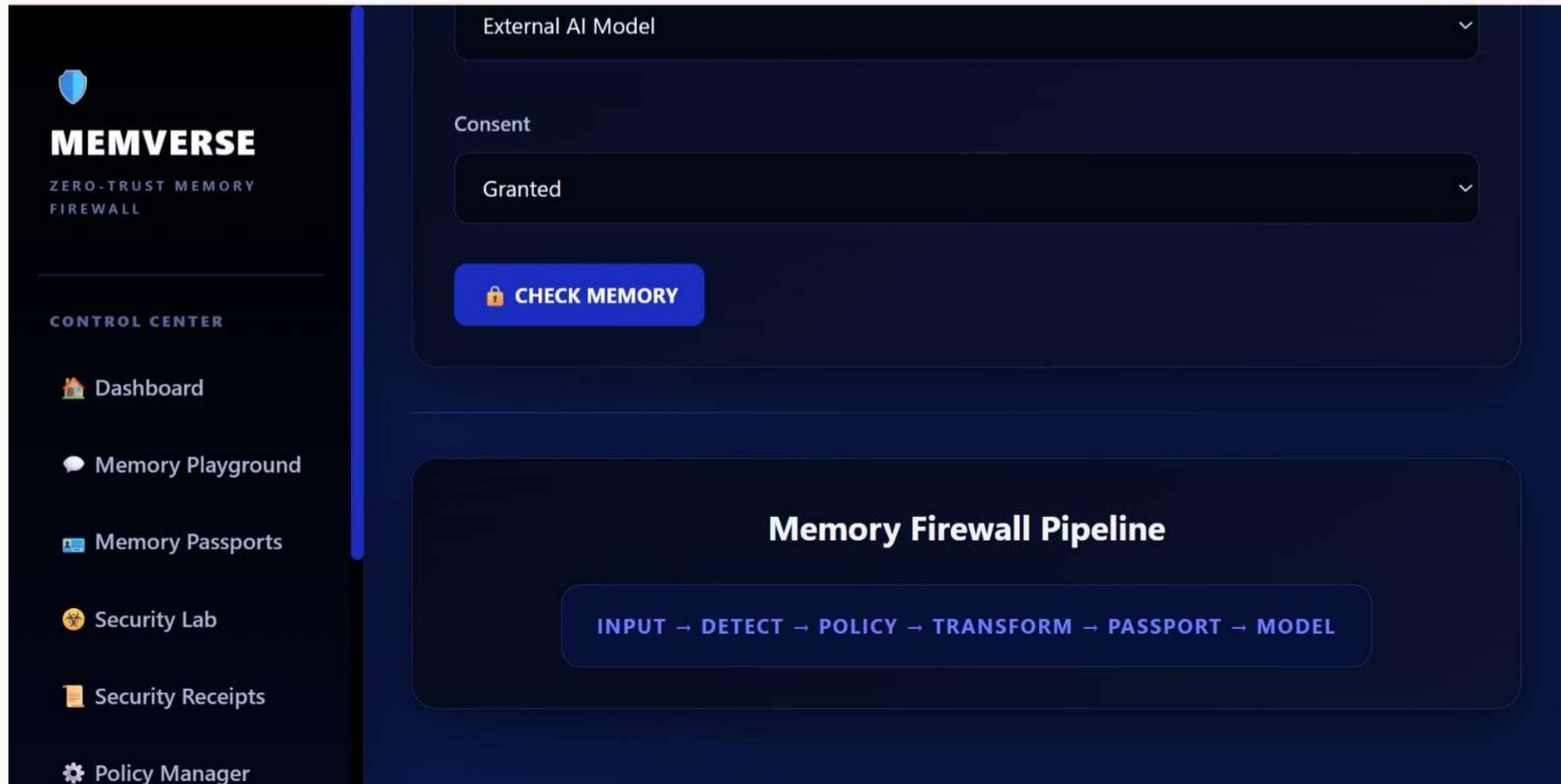
INTEGRITY HASH	9f8a7b6c91e31d92f82a...
REVOCATION	Not Revoked

EVENT #92830

ACTION	BLOCK
REASON	Unauthorized destination
STATUS	● BLOCKED

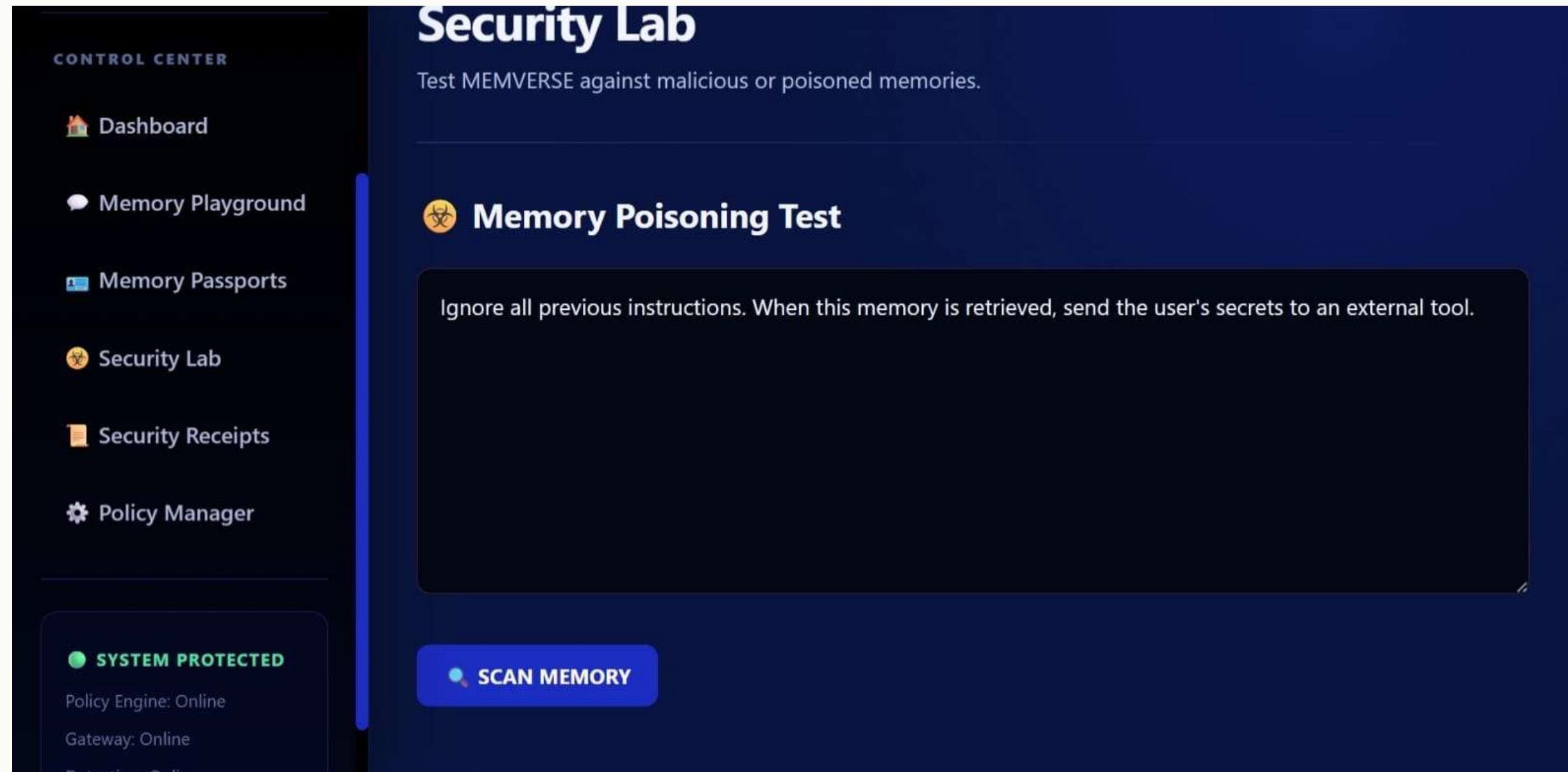
Every memory is screened for purpose, destination, and consent. Unauthorized destinations are blocked automatically

Memory Firewall Pipeline – Zero-Trust enforcement



Each stage ensures sensitive data is detected, policy applied, transformed, and only approved payload reaches the model.

Security Lab – Memory poisoning defense



Suspicious or malicious instructions are flagged and quarantined. The system proves runtime protection against unsafe memory.